In [1]:
```
from sklearn import datasets
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import seaborn as sns
```

In [2]:
```
iris = datasets.load_iris()

df = pd.DataFrame(iris['data'], columns=iris['feature_names'])
df.head()
```

Out[2]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

In [3]:
```
scaler = StandardScaler()

scaled_data = pd.DataFrame(scaler.fit_transform(df))

scaled_data
```

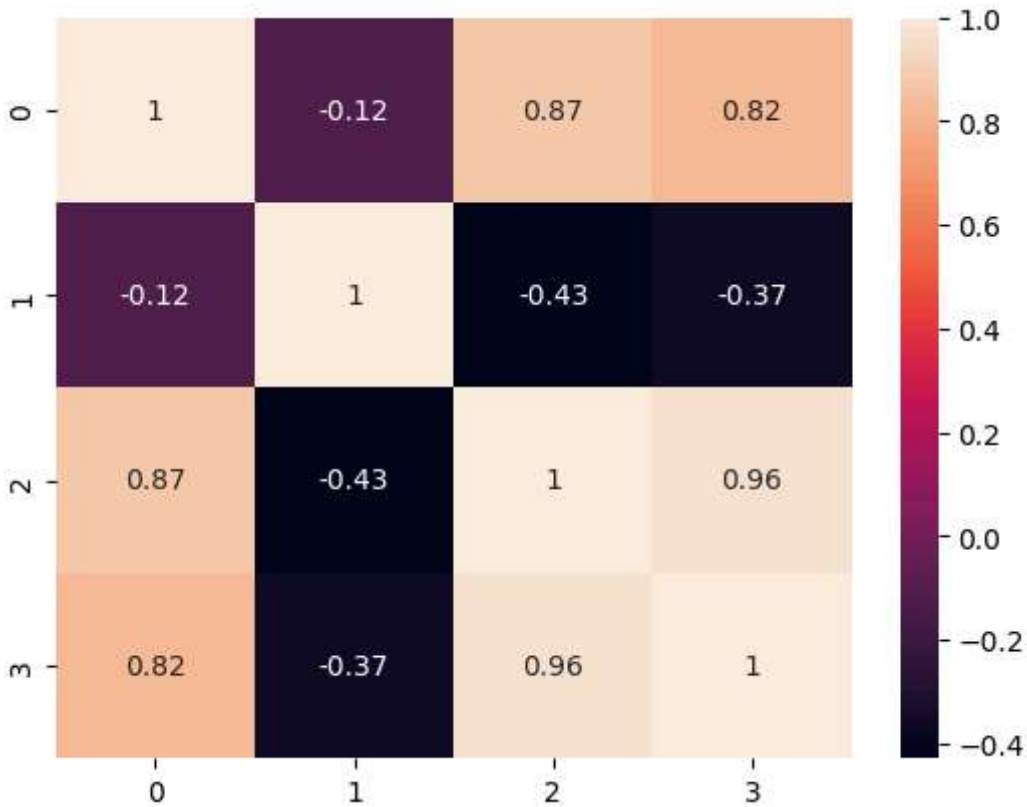
Out[3]:

	0	1	2	3
0	-0.900681	1.019004	-1.340227	-1.315444
1	-1.143017	-0.131979	-1.340227	-1.315444
2	-1.385353	0.328414	-1.397064	-1.315444
3	-1.506521	0.098217	-1.283389	-1.315444
4	-1.021849	1.249201	-1.340227	-1.315444
...	...	...	...	...
145	1.038005	-0.131979	0.819596	1.448832
146	0.553333	-1.282963	0.705921	0.922303
147	0.795669	-0.131979	0.819596	1.053935
148	0.432165	0.788808	0.933271	1.448832
149	0.068662	-0.131979	0.762758	0.790671

150 rows × 4 columns

In [4]:
```
sns.heatmap(scaled_data.corr(), annot=True)
```

Out[4]: <Axes: >



In [5]:
```
pca = PCA(n_components=3)

pca.fit(scaled_data)
```

Out[5]:
```
▾ PCA
▸ Parameters
```

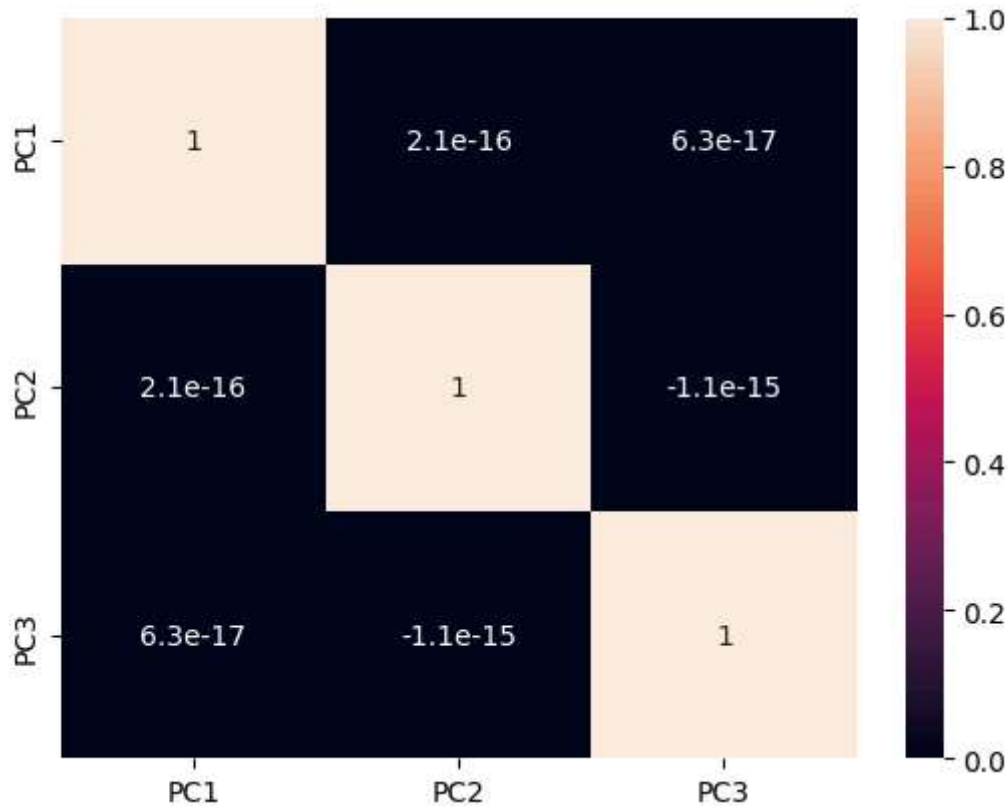
In [6]:
```
data_pca = pca.transform(scaled_data)

data_pca = pd.DataFrame(data_pca, columns=['PC1','PC2','PC3'])

data_pca.head()
```

Out[6]:

	PC1	PC2	PC3
0	-2.264703	0.480027	0.127706
1	-2.080961	-0.674134	0.234609
2	-2.364229	-0.341908	-0.044201
3	-2.299384	-0.597395	-0.091290
4	-2.389842	0.646835	-0.015738

In [7]:
```
sns.heatmap(data_pca.corr(), annot=True)
```

Out[7]: <Axes: >



In [ ]: